Module 2

HTML Fundamentals

Agenda

1. HTML Document Structure
2. Elements, Tags and Attributes
3. Headings and Paragraphs
4. Text Formatting
5. Lists
6. Links and Navigation
7. Images
8. Tables
9. Audio and Video Elements
10. HTML Comments and Code Organization

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HTML Document Structure

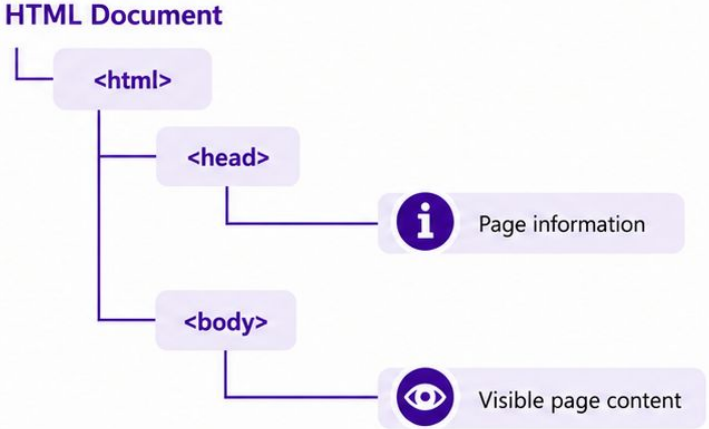
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Understanding an HTML Document

HTML (HyperText Markup Language) is used to define the **structure** and **content** of a web page.

An HTML document is organized into a **hierarchy of elements** that tells the browser what each part of the page represents.

Basic Structure



Two Main Sections

1. **<head>**
   Contains metadata and information about the page (not visible on the browser window).

2. **<body>**
   Contains the actual content that is visible on the web page.

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Basic HTML Document Structure

Every HTML5 document follows a standard structure that helps the **browser understand** and **display** the **webpage correctly**.

Standard HTML5 Structure

```
<!DOCTYPE html>
<html>
   <head>
      <title>My Web Page</title>
   </head>
   <body>
      <h1>Welcome to My Web Page</h1>
      <p>This is my first webpage.</p>
   </body>
</html>
```

Component	Purpose
`<!DOCTYPE html>`	Declares that the document uses HTML5.
`<html>`	Root element containing the entire document.
`<head>`	Contains document information and metadata.
`<title>`	Defines the title shown in the browser tab.
`<body>`	Contains the visible content of the webpage.

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Elements, Tags and Attributes

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HTML Tags and Elements

HTML uses **tags and elements** to define the structure and content of a webpage.

What Is a Tag?

A **tag** is a markup keyword enclosed in angle brackets `<>`.

Example:

```
<h1>
```

Most HTML elements use an **opening tag** and a **closing tag.**

```
<h1>Welcome</h1>
```

- `<h1>` → Opening tag
- `</h1>` → Closing tag

What Is an Element?

An **HTML element** generally consists of:

Opening Tag + Content + Closing Tag

Example:

```
<p>Hello, World!</p>
```

`<p>`	→	Opening tag
Hello, World!	→	Content
`</p>`	→	Closing tag

Together → Complete HTML element

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Void Elements

Some HTML elements do not have a closing tag because they **cannot contain content.**

These are called **void elements.**

Examples of Void Elements

Element	Purpose	Example Usage
` `	Inserts a line break	`Hello World`
`<hr>`	Inserts a horizontal rule (thematic break)	`<hr>`
`<img>`	Embeds an image	`<img src="image.jpg" alt="Sample">`
`<input>`	Creates an input field	`<input type="text" name="username">`

Code Example

index.html

```
<!DOCTYPE html>
<html>
<body>
    Hello<br>World!
    <hr>
    <img src="image.jpg" alt="Sample Image">
    <input type="text" placeholder="Enter name">
</body>
</html>
```

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HTML Attributes

HTML attributes provide additional information about an HTML element and can control its behavior or identify it.

Attribute Syntax

Attributes are written **inside the opening tag.**

```
<tag attribute="value">
```

Example

```
<a href="https://example.com">Visit Website</a>
```

Here:

- `<a>` → HTML tag
- `href` → Attribute
- `"https://example.com"` → Attribute value
- `Visit Website` → Element content

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Common Attributes

Attribute	Purpose	Example
`id`	Uniquely identifies an element	`id="header"`
`class`	Groups elements for styling/selection	`class="box"`
`title`	Provides additional information	`title="More information"`
`href`	Specifies a link destination	`href="about.html"`
`src`	Specifies a resource location	`src="image.jpg"`
`alt`	Provides alternative text for an image	`alt="Company logo"`

Important Rules

- Attributes are placed in the **opening tag.**
- The attribute value is normally written inside **quotation marks.**
- Multiple attributes can be used in one element.

Example

```
<img src="logo.png" alt="Company Logo"
 title="Our Logo">
```

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Id & Class

id Attribute

The id **attribute** gives an HTML element a **unique name or identifier.** It is mainly used when you need to target **one specific element** on a webpage.

- An id should normally be **unique within a page**.
- The same id should not be assigned to multiple elements.
- It is commonly used with **CSS, JavaScript, and links**.
- An id value should not contain spaces.

Example HTML

```
<h1 id="main-heading">Welcome to My Website</h1>
```

Here, `main-heading` uniquely identifies the `<h1>` element.

You can target it using CSS:

CSS

```
#main-heading {
   color: blue;
}
```

Output

Welcome to My Website

class Attribute

The **class attribute** assigns an element to a **group or category.** It is useful when the **same styling or behavior needs to be applied to multiple elements.**

- The same class can be used on **multiple elements**.
- An element can have **more than one class**.
- It is commonly used with **CSS and JavaScript**.
- Class names should be meaningful and should not contain spaces within a single class name.

Example HTML

```
<p class="highlight">Important information</p>
<p class="highlight">Read this carefully</p>
```

Both paragraphs belong to the `highlight` class.

You can target them using CSS:

CSS

```
.highlight {
   background-color: yellow;
}
```

Output

Important information

Read this carefully

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Elements, Tags and Attributes Together

HTML **tags, elements, and attributes** work together to define and provide information about **webpage** content.

Example

```
<a href="https://example.com" title="Visit Example">
 Visit Website
</a>
```



Clicking this link takes the user to https://example.com

Understanding the Example

`<a>`	→	Opening tag
`href="https://example.com"`	→	Attribute
`title="Visit Example"`	→	Attribute
`Visit Website`	→	Content
`</a>`	→	Closing tag
Complete structure	→	HTML element

How It All Works Together

Tag
Defines the element

→

Attributes
Provide additional information

→

Content
Displayed to the user

→

Complete Element
All parts work together

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Example

Let's see how different **HTML attributes** are used in a webpage.

HTML Code

```
1   <!DOCTYPE html>
2   <html>
3   <head>
4      <title>HTML Attributes Example</title>
5   </head>
6   <body>
7      <h1 id="header">Welcome to My Website</h1>
8      <p class="intro">This is the first paragraph.</p>
9      <p class="box">This is the second paragraph.</p>
10     <p title="This is additional information">
11       Hover over this text
12     </p>
13     <a href="https://example.com">Visit Website</a>
14     <img src="image.jpg" alt="Company Logo">
15     <img src="logo.png" alt="Company Logo" title="Our Logo">
16  </body>
</html>
```

What Each Attribute Does

1. **id attribute**
   Uniquely identifies an element.
   Used here: `id="header"` in `<h1>`
2. **class attribute**
   Groups elements for styling or selection.
   Used here: `class="intro"` and `class="box"`
3. **title attribute**
   Provides additional information when you hover over the element.
4. **href attribute**
   Specifies where the link should go.
   Used here: `href="https://example.com"`
5. **src attribute**
   Specifies the location/source of an image.
   Used here: `src="image.jpg"` and `src="logo.png"`
6. **alt attribute**
   Provides alternative text if the image cannot be displayed.

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HTML Syntax Pattern

HTML Syntax Pattern

```
<element attribute="value">
   Content
</element>
```

Important Rules

- Tags are enclosed in < >.
- Closing tags use /.
- Attributes are written inside the opening tag.
- Attribute values are placed inside quotation marks.
- Elements can be **nested** inside other elements.

Nested Example

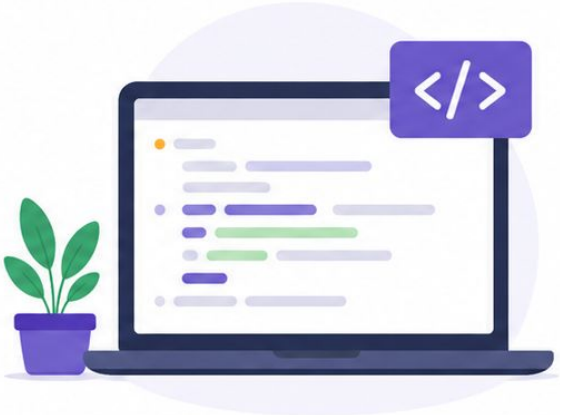
```
<div>
    <h1>My Website</h1>
    <p>Welcome to my website.</p>
</div>
```

Here, the <h1> and <p> elements are **nested inside** the element.

Output (Preview in Browser)

My Website

Welcome to my website.



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Headings and Paragraphs

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Headings and Paragraphs

HTML provides **headings** and **paragraphs** to organize and present textual content on a webpage.

Headings

HTML provides **six levels** of headings.

```
<h1>Main Heading</h1>
<h2>Section Heading</h2>
<h3>Subsection Heading</h3>
<h4>Heading Level 4</h4>
<h5>Heading Level 5</h5>
<h6>Heading Level 6</h6>
```

- <h1> → Highest-level heading
- <h6> → Lowest-level heading
- Headings should follow a **logical hierarchy**

Paragraphs

The <p> element defines a paragraph.

```
<p>This is a paragraph of text.</p>
<p>This is another paragraph.</p>
```

Each <p> element represents a separate paragraph.

Example

```
<h1>Web Development</h1>
<h2>HTML</h2>
<p>HTML defines the structure of a web page.</p>
<h2>CSS</h2>
<p>CSS controls the presentation of a web page.</p>
```

Preview in Browser

Web Development

HTML

HTML defines the structure of a web page.

CSS

CSS controls the presentation of a web page.

Key Takeaway:

Use headings for titles and subtitles, and paragraphs for textual content to create clear and structured webpages.

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Text Formatting

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Basic Text Formatting

HTML provides elements for **emphasizing, highlighting, and presenting text** with different meanings or visual appearances.

Common Text Formatting Elements

Element	Purpose	Example	
`<strong>`	Indicates **strong importance**	`<strong>This is important text.</strong>`	This is **important text.**
`<em>`	Indicates **emphasis**	`<em>This is emphasized text.</em>`	*This is emphasized text.*
`<b>`	Draws **attention** to text without adding importance	`<b>This text is bold.</b>`	**This text is bold.**
`<i>`	Represents text in an **alternate voice or style**	`<i>This text is italic.</i>`	*This text is italic.*
`<u>`	Represents **underlined** text	`<u>This text is underlined.</u>`	This text is underlined.
`<mark>`	**Highlights** text	`<mark>This text is highlighted.</mark>`	This text is highlighted.

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Example

Let's see how HTML text formatting elements appear in a browser.

HTML Code

```
1  <body>
2    <p>
3      <strong>Important:</strong> Submit the
4      assignment today.
5    </p>
6
7    <p>
8      This word is <em>emphasized</em>.
9    </p>
10
11   <p>
12     This is <b>bold</b> and this is <i>italic</i>.
13   </p>
14   <p>
15     This is <u>underlined</u> and this is
       <mark>highlighted</mark>.
16 </body>
```

Output (In Browser)

Important: Submit the assignment today.

This word is *emphasized*.

This is **bold** and this is *italic*.

This is underlined and this is highlighted.

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Additional Text Elements

HTML also provides elements for **smaller text**, **scientific notation**, **inserted/deleted content**, and **line or section breaks**.

Element	Purpose
`<small>`	Represents **smaller** or less prominent text
`<sub>`	Displays **subscript** text
`<sup>`	Displays **superscript** text
`<del>`	Represents **deleted** content
`<ins>`	Represents **inserted** content
` `	Creates a **line break**
`<hr>`	Represents a **thematic break** between sections

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Examples

Let's see how additional text elements appear in a browser.

Small Text

```
<small>Terms and conditions apply.</small>
```

Subscript and Superscript

```
H<sub>2</sub>O
x<sup>2</sup>
```

Deleted and Inserted Content

```
<del>₹500</del>
<ins>₹400</ins>
```

Line Break

```
Hello<br>
World
```

Horizontal/Thematic Break

```
<hr>
```

Output (In Browser)

Terms and conditions apply.

H_2O

x^2

~~₹500~~ ₹400

Hello
World

Chapter 1

Introduction to HTML

Chapter 2

Getting Started

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Lists

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Unordered and Ordered Lists

HTML provides lists to **organize related items** in a clear and structured way

Unordered List

An **unordered list** displays items using bullets.

- `<ul>` → Defines the unordered list
- `<li>` → Defines each list item

Example Code:

HTML

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

Result:

- HTML
- CSS
- JavaScript

Output (In Browser)

- HTML
- CSS
- JavaScript

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Unordered and Ordered Lists

Ordered List

An **ordered list** displays items in a sequence, usually using numbers.

- `<ol>` → Defines the ordered list
- `<li>` → Defines each list item



Example Code:

HTML

```
<ol>
   <li>Open VS Code</li>
   <li>Create an HTML file</li>
   <li>Write HTML code</li>
</ol>
```

Result:

1. Open VS Code
2. Create an HTML file
3. Write HTML code

Output (In Browser)

1. Open VS Code
2. Create an HTML file
3. Write HTML code



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Nested and Description Lists

HTML allows lists to be **nested** and also provides **description lists** for presenting terms and their descriptions.

Nested Lists

A nested list is a list placed inside another list item.



HTML Code

```
<ul>
  <li>Frontend
    <ul>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript</li>
    </ul>
  </li>
  <li>Backend
    <ul>
      <li>Python</li>
      <li>Java</li>
    </ul>
  </ul>
```



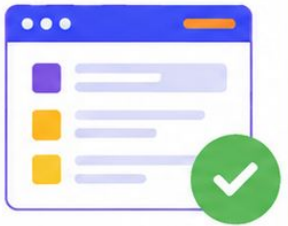
Structure:



- Frontend
  - HTML
  - CSS
  - JavaScript
- Backend
  - Python
  - Java

Output (In Browser)

- Frontend
  - HTML
  - CSS
  - JavaScript
- Backend
  - Python
  - Java



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Nested and Description Lists

Description Lists

A description list is used to associate terms with their descriptions.

- `<dl>` → Description list
- `<dt>` → Term
- `<dd>` → Description



Example Code:

HTML

```
<dl>
    <dt>HTML</dt>
    <dd>Defines the structure of a webpage.</dd>

    <dt>CSS</dt>
    <dd>Controls the presentation of a webpage.</dd>
</dl>
```

Output (In Browser)

HTML

Defines the structure of a webpage.

CSS

Controls the presentation of a webpage.

When to Use

- **Nested lists** → Organize hierarchical information
- **Description lists** → Present terms, definitions, or key-value information



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Hands-On

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Hands-On: Create a Skills & Technologies List

Objective

Use HTML lists to organize information.

Task

Create a webpage section called: **My Skills**

1 Part A — Unordered List

Create a list containing:

- HTML
- CSS
- JavaScript

2 Part B — Ordered List

Create an ordered list showing:

1. Open VS Code
2. Create an HTML file
3. Write HTML code
4. Open the webpage in a browser

3 Part C — Nested List

Create a nested list:

- Frontend
  - HTML
  - CSS
  - JavaScript
- Backend
  - Python
  - Java

4 Part D — Description List

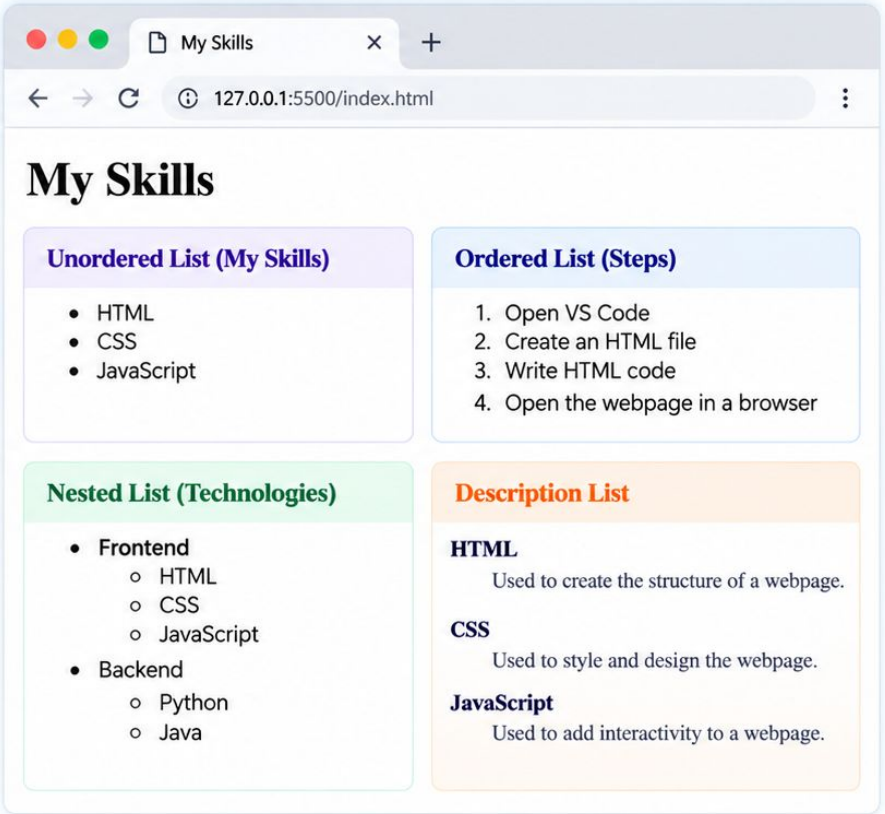
Create a description list for:

- HTML
- CSS
- JavaScript

Provide a short description for each.

Expected Output

All four types of lists should be visible correctly in the browser.



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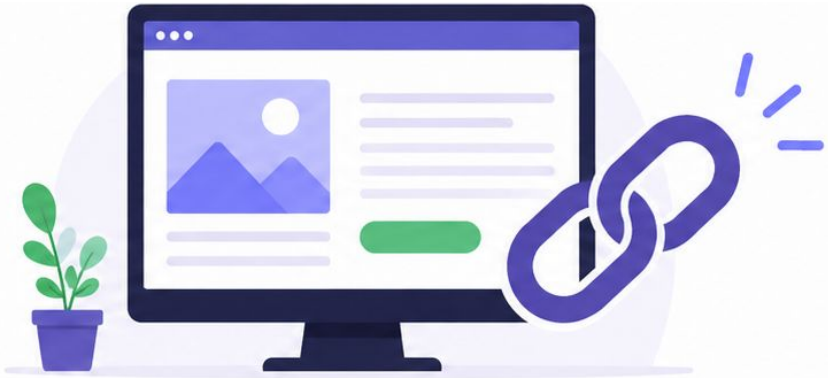
Links and Navigation

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Creating Links

HTML links allow users to **navigate from one webpage or resource to another**.

The `<a>` **(anchor)** element is used to create a link.



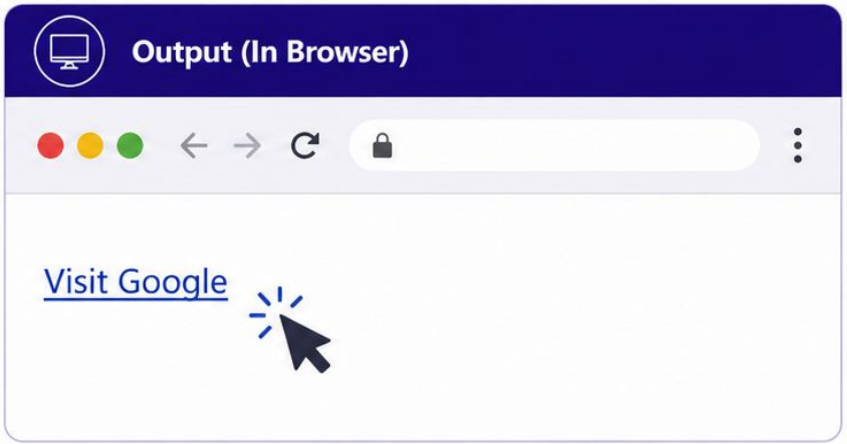
Basic Syntax

```
<a href="URL">Link Text</a>
```

- `<a>` → Defines a hyperlink
- `href` → Specifies the destination
- `Link Text` → Text displayed to the user

Example — External Link

```
<a href="https://www.google.com">Visit Google</a>
```



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Absolute vs Relative URLs

Absolute URL — Points to a complete web address.

```
<a href="https://example.com/about">About Us</a>
```

Relative URL — Points to a resource within the same website.

```
<a href="about.html">About Us</a>
```

Opening a Link in a New Tab

The `target="_blank"` attribute opens the linked page in a new browsing context, commonly a new tab.

```
<a href="https://example.com"
    target="_blank">
    Visit Example
</a>
```



Output (In Browser)

https://yourwebsite.com

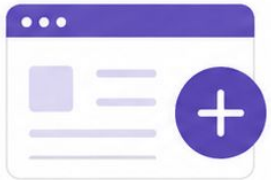
About Us (Navigates to https://example.com/about)

About Us (Navigates to about.html within the site)

Output (In Browser)

https://yourwebsite.com

Visit Example (Opens in a new tab)



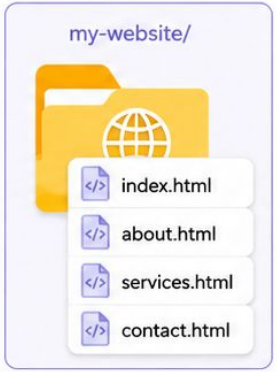
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Navigation Between Web Pages

Navigation allows users to move between different pages of the same website using links.

Example Website Structure

- my-website/
  - index.html
  - about.html
  - services.html
  - contact.html

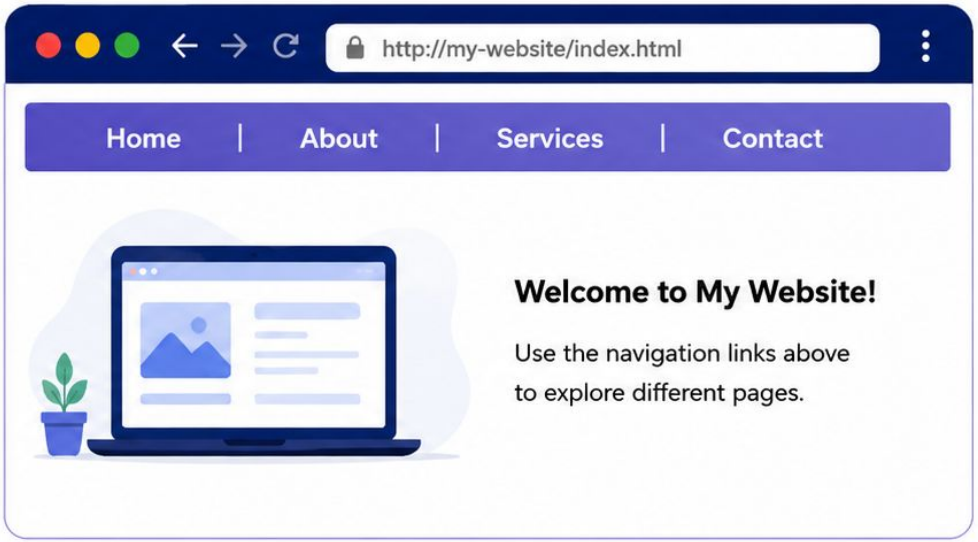


Creating Navigation Links

Inside index.html:

```
<nav>
   <a href="index.html">Home</a>
   <a href="about.html">About</a>
   <a href="services.html">Services</a>
   <a href="contact.html">Contact</a>
</nav>
```

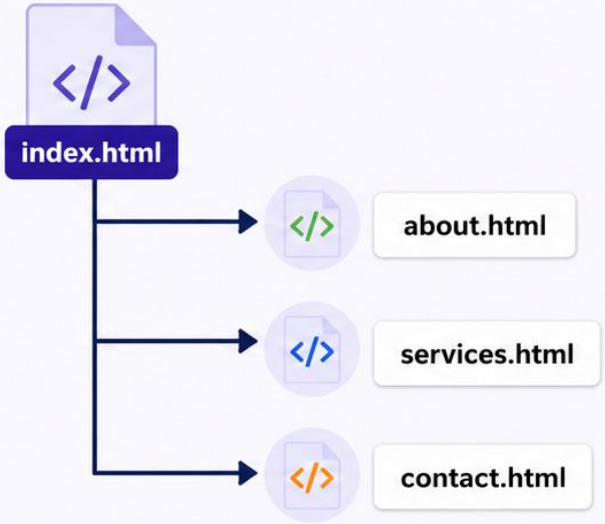
Output (In Browser)



Key Takeaway: Navigation links help users move easily between pages within the same website.

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How It Works





When a user clicks a link, the browser loads the corresponding **HTML page**.

Why Navigation Is Important

- Connects different pages of a website
- Helps users move between sections
- Provides a consistent way to explore website content
- Forms the basic structure of a multi-page website

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Hands-On

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Hands-On: Create Links & Navigation

1 Project Structure

Create the following files:



2 Add Navigation Links

Add navigation links to each page.

```
<a href="index.html">Home</a>
<a href="about.html">About</a>
<a href="services.html">Services</a>
<a href="contact.html">Contact</a>
```

3 Add an External Link

Add a link to an external website using an absolute URL.

```
<a href="https://example.com">
   Visit Website
</a>
```

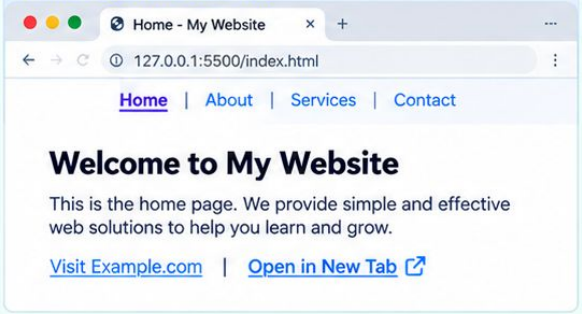
4 Open a Link in a New Tab

Use `target="_blank"` to open a link in a new tab.

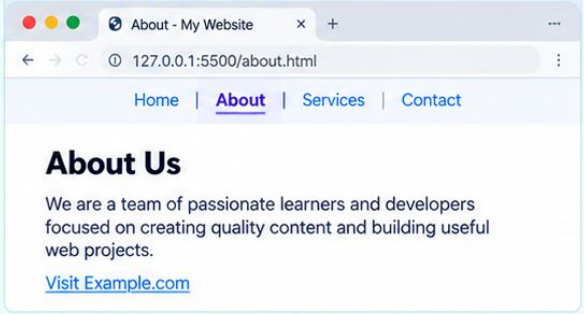
```
<a href="https://example.com" target="_blank">
   Visit Website (New Tab)
</a>
```

Sample Output (Web Pages)

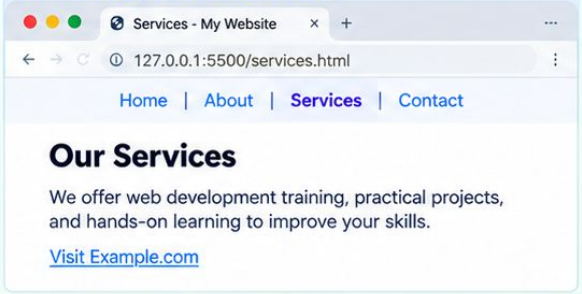
Home Page (index.html)



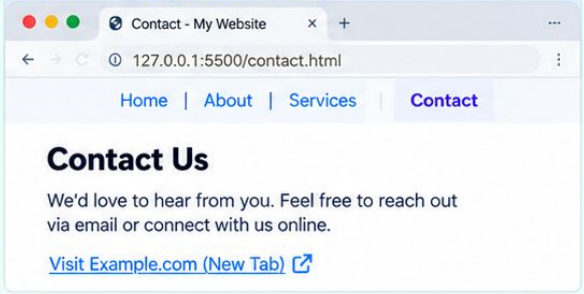
About Page (about.html)



Services Page (services.html)



Contact Page (contact.html)



Expected Outcome

You should be able to navigate between Home → About → Services → Contact and back again. One link should open an external website, and one link should open in a new tab.

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Images

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Adding Images to a Web Page

HTML allows us to add images to a webpage using the `<img>` element.

Basic Syntax

```
<img src="image.jpg" alt="Description of the image">
```

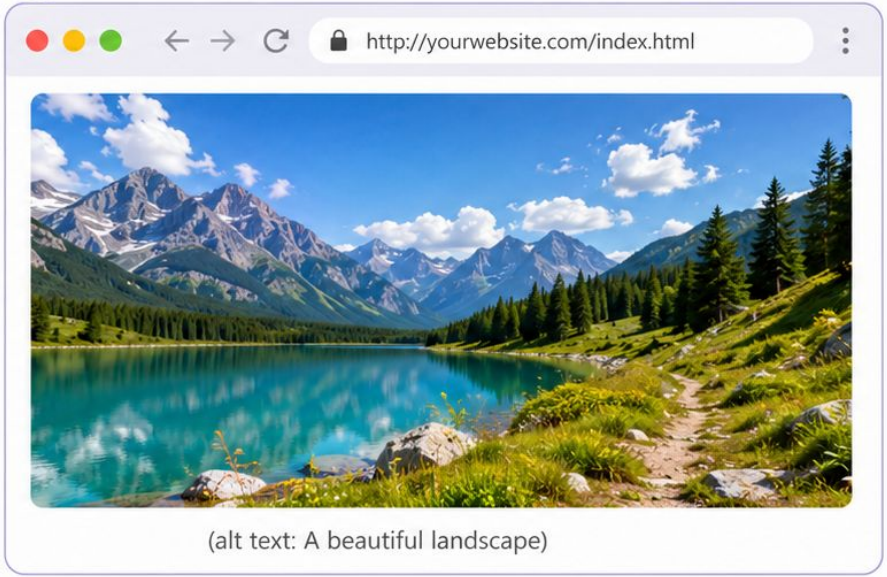
- `<img>` → Defines an image
- `src` → Specifies the image location
- `alt` → Provides alternative text describing the image

Example

```
<img src="image.jpg" alt="A beautiful landscape">
```

Tip: Always use meaningful `alt` text for accessibility and better SEO.

Output (In Browser)





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Adding Images to a Web Page

Example — Local Image

Suppose the project contains:



HTML Code:

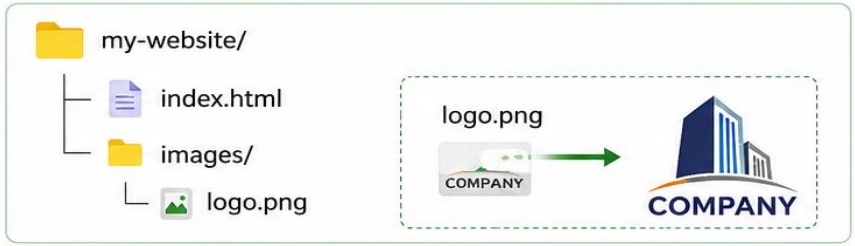
```
<img src="image.jpg" alt="A beautiful landscape">
```

We can display the image using:

Output (In Browser)



Using an Image from a Folder



HTML Code:

```
<img src="images/logo.png" alt="Company logo">
```

Why alt Is Important

- **Describes the image** when it cannot be displayed
- **Helps users using screen readers** understand the image
- **Provides meaningful alternative content**

the

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Image Size and Accessibility

HTML provides attributes that help control **image dimensions** and make images more accessible to users.

Controlling Image Size

The `width` and `height` attributes can specify the displayed dimensions of an image.

```
<img
  src="logo.png"
  alt="Company logo"
  width="300"
  height="150">
```

- `width` → Specifies the displayed width
- `height` → Specifies the displayed height

Output (In Browser)



Setting image dimensions ensures consistent layout and improves page load experience.

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Writing Meaningful alt Text

The `alt` attribute should describe the **purpose** or **content** of the image.

Good:
Descriptive `alt` text that explains the image.

```
<img src="student.jpg" alt="Student working on a laptop">
```

Poor:
Unhelpful alt text that does not describe the image.

```
<img src="student.jpg" alt="image">
```

Meaningful `alt` text improves **accessibility** for users who rely on screen readers and enhances **SEO**.

Common Mistakes

- Incorrect image file path
- Missing `alt` attribute
- Using meaningless `alt` text
- Using unnecessarily large image dimensions

Output (In Browser)

Good (Descriptive alt Text)



If the image cannot be displayed, screen readers will read: **"Student working on a laptop"**

Poor (Unhelpful alt Text)



If the image cannot be displayed, screen readers will read: **"image"**

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Hands-On

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Hands-On: Add and Customize Images

1 Add an Image

Use the <img> element to add an image.

```
<img src="image.jpg" alt="A beautiful landscape">
```

2 Use an Image from a Folder

Create the following project structure:



Update the image path:

```
<img src="images/landscape.jpg" alt="A beautiful landscape">
```

3 Set Image Size

Set the image dimensions using width and height.

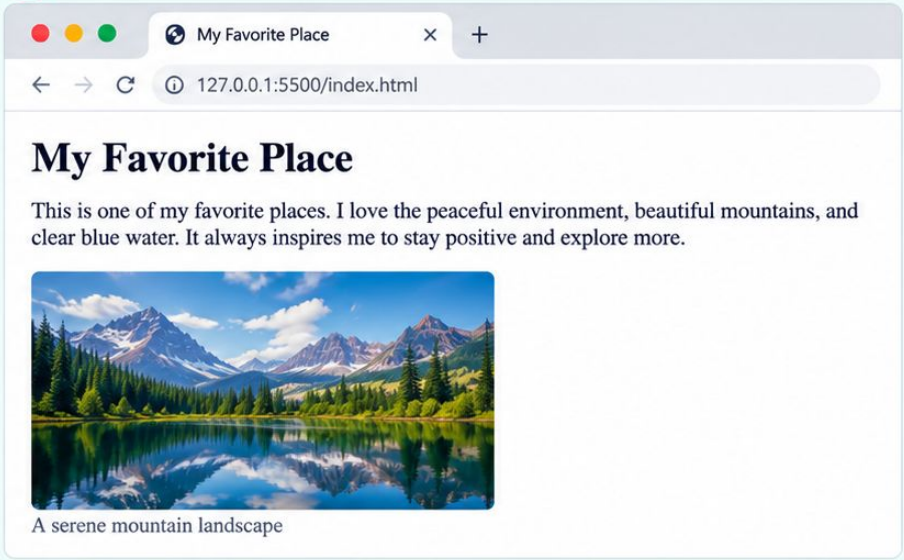
```
<img src="images/landscape.jpg"
     alt="A beautiful landscape"
     width="300"
     height="150">
```

4 Write Meaningful alt Text

Use descriptive alt text that explains the image.

```
<img src="images/landscape.jpg" alt="A serene mountain landscape">
```

Expected Output (How the webpage should look)



Challenge

1. Change the image width and height.
2. Give the image a meaningful alt description.
3. Temporarily enter an incorrect image path and observe what happens.
4. Fix the path and verify that the image displays correctly.

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Tables

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HTML Tables

Core Idea

An **HTML table** is used to organize and display data in **rows** and **columns**.

Main Table Elements

Element		Purpose
`<table>`		Defines the table
`<tr>`		Defines a table row
`<th>`	Abc	Defines a header cell
`<td>`	Abc	Defines a data cell

Example Code

```
<table border="1">
  <tr>
    <th>Student</th>
    <th>Course</th>
    <th>Marks</th>
  </tr>
  <tr>
    <td>Alice</td>
    <td>HTML</td>
    <td>95</td>
  </tr>
  <tr>
    <td>Bob</td>
    <td>CSS</td>
    <td>88</td>
  </tr>
</table>
```

Output (In Browser)

Student	Course	Marks
Alice	HTML	95
Bob	CSS	88

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Basic Table Structure

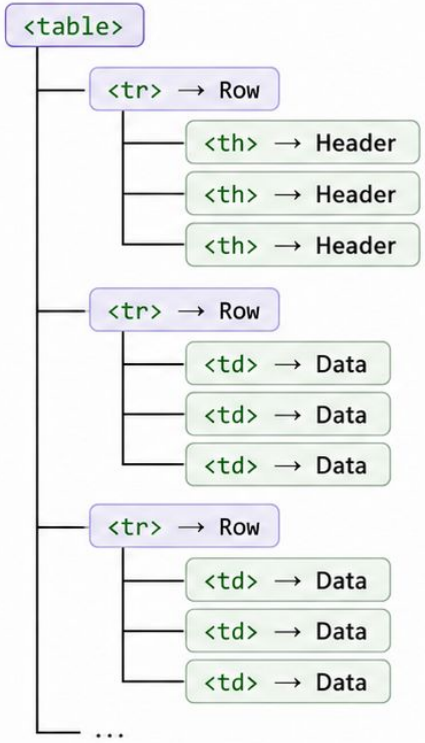
HTML Code

```
<table>
    <tr>
        <th>Name</th>
        <th>Age</th>
        <th>City</th>
    </tr>

    <tr>
        <td>Rahul</td>
        <td>21</td>
        <td>Delhi</td>
    </tr>

    <tr>
        <td>Priya</td>
        <td>22</td>
        <td>Mumbai</td>
    </tr>
</table>
```

Table Structure



Output (In Browser)

Name	Age	City
Rahul	21	Delhi
Priya	22	Mumbai

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Table Structure and Spanning

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For larger tables, HTML provides elements to **organize table sections** and attributes to **merge cells across rows or columns**.

Organizing Table Sections

<thead>
→ Contains the table **header rows**

<tbody>
→ Contains the main **table data**

<tfoot>
→ Contains summary or **footer rows**

Example

```
<table border="1">
  <thead>
    <tr>
      <th>Product</th>
      <th>Price</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Laptop</td>
      <td>₹50,000</td>
    </tr>
    <tr>
      <td>Phone</td>
      <td>₹20,000</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td><strong>Total:</strong></td>
      <td><strong>₹70,000</strong></td>
    </tr>
  </ftoot>
</table>
```

Output (In Browser)

Product	Price
Laptop	₹50,000
Phone	₹20,000
Total:	**₹70,000**

Use `thead`, `tbody`, and `tfoot` to structure your table for **better readability and accessibility.**

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Spanning Table Cells

colspan merges cells across columns:

```
<td colspan="2">Total</td>
```

rowspan merges cells across rows:

```
<td rowspan="2">Electronics</td>
```

Output (In Browser)

Total	
Product	**Price**
Laptop	₹50,000
Phone	₹20,000
Electronics	**Total Price**
	₹70,000

</> Example Code

```
<table border="1" cellspacing="0" cellpadding="8">
  <tr>
    <td colspan="2"><strong>Total</strong></td>
  </tr>
  <tr>
    <th>Product</th>
    <th>Price</th>
  </tr>
  <tr>
    <td>Laptop</td>
    <td>₹50,000</td>
  </tr>
  <tr>
    <td>Phone</td>
    <td>₹20,000</td>
  </tr>
  <tr>
    <td rowspan="2"><strong>Electronics</strong></td>
    <td><strong>Total Price</strong></td>
  </tr>
  <tr>
    <td>₹70,000</td>
  </tr>
</table>
```

Merges across 2 columns

Merges across 2 rows

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Hands-On

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Hands-On: Create an HTML Table

1 Objective

Create an HTML table to organize information into rows and columns.

2 Student Information Table

Create a table with the following data:

Name	Age	City
Rahul	21	Delhi
Priya	22	Mumbai
Aman	20	Pune
Neha	21	Bangalore

3 Use the Following Tags

- `<table>` – Table
- `<tr>` – Table row
- `<th>` – Header cell
- `<td>` – Data cell

Use `<thead>` for the header, `<tbody>` for the data, and `<tfoot>` for the footer.

4 Add a Footer

Add a `<tfoot>` section containing:

```
Total Students: 4
```

5 Practice Cell Spanning

- Use `colspan` to merge cells across columns.
- Use `rowspan` to merge cells across rows.

Expected Output (How the webpage should look)

Student Information

Name	Age	City
Rahul	21	Delhi
Priya	22	Mumbai
Aman	20	Pune
Neha	21	Bangalore
Total Students: 4		

Product Table (Using Colspan & Rowspan)

Product		Price	Category
Electronics	Laptop	₹50,000	
	Smartphone	₹25,000	
	Headphones	₹2,000	
Books	Java Programming	₹800	
	Web Development	₹600	
Total Products: 5			

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Audio and Video Elements

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Adding Audio to a Web Page

HTML allows us to embed **audio content** directly into a webpage using the `<audio>` element.

Basic Syntax

```
<audio controls>
    <source src="music.mp3"
            type="audio/mpeg">
</audio>
```

- `<audio>` → Defines an audio player
- `controls` → Displays play, pause, volume, and other controls
- `<source>` → Specifies the audio file and its format
- `src` → Specifies the audio file location
- `type` → Specifies the audio format

Output (In Browser)



Good to Know

- Use audio files in formats like MP3, OGG, or WAV.
- Always provide the correct type for better compatibility.
- Audio content should enhance the user experience without auto-playing unexpectedly.

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Example Project Structure

Project Structure

my-website/

- index.html
- music.mp3

Single Audio Source

```
<audio controls>
   <source src="music.mp3" type="audio/mpeg">
</audio>
```

Output (In Browser)



Audio player with controls

Multiple Audio Sources

Different browsers may support different audio formats, so multiple sources can be provided:

```
<audio controls>
   <source src="music.mp3" type="audio/mpeg">
   <source src="music.ogg" type="audio/ogg">
   Your browser does not support audio.
</audio>
```

How it works

The browser will use the first supported format and ignore the others.

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Adding Video to a Web Page

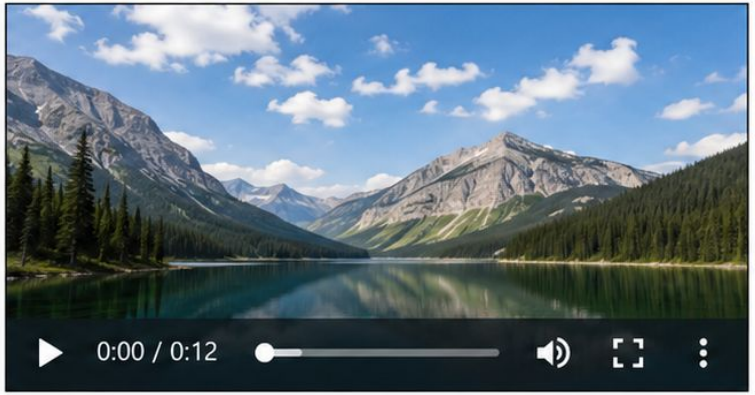
HTML allows us to embed **video content** directly into a webpage using the `<video>` element.

Basic Syntax

```
<video controls width="640">
   <source src="video.mp4" type="video/mp4">
</video>
```

- `<video>` → Defines a video player
- `controls` → Displays play, pause, volume, and other controls
- `width` → Specifies the displayed width
- `<source>` → Specifies the video file and format
- `src` → Specifies the video file location
- `type` → Specifies the video format

Output (In Browser)



Good to Know

- Use formats like MP4, WebM, or OGG for better browser support.
- Always set meaningful `width` and controls for a better user experience.
- Provide multiple `<source>` elements for maximum compatibility.

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Example Project Structure

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Project Structure

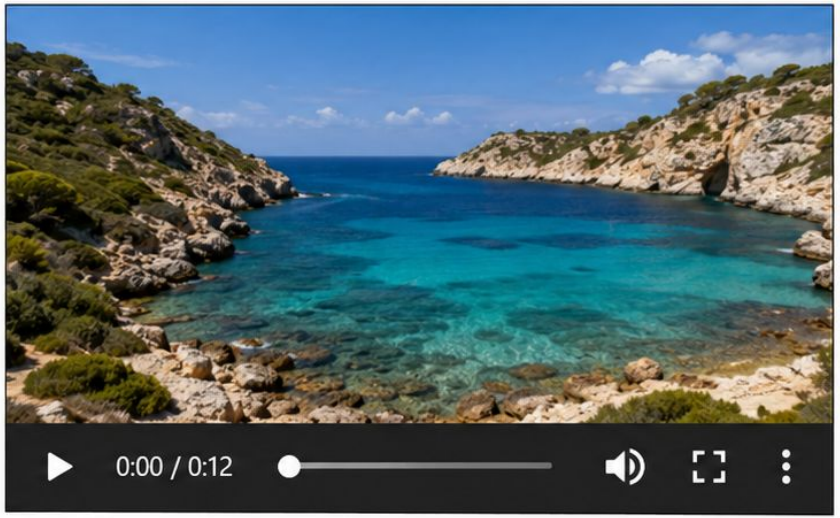
my-website/

- index.html
- video.mp4

HTML Code

```
<video controls width="640">
    <source src="video.mp4"
            type="video/mp4">
</video>
```

Output (In Browser)



Tip: Place video files in the project folder and use the correct path in the `<source>` element.

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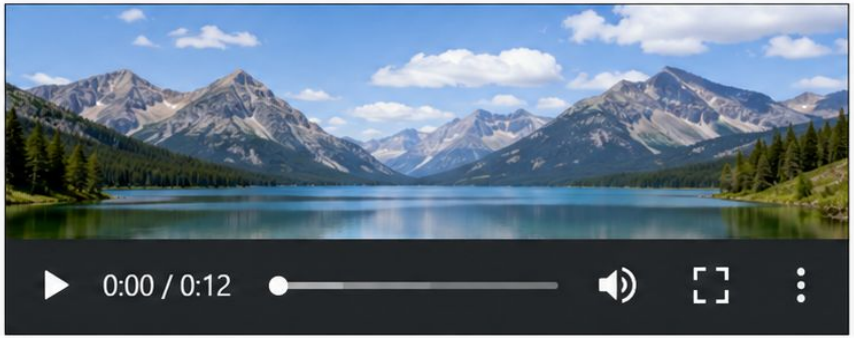
Useful Video Attributes

Attribute	Purpose
controls	Shows playback controls
width	Sets displayed width
height	Sets displayed height
poster	Displays an image before the video plays
autoplay	Starts playback automatically
loop	Repeats the video

Multiple Video Sources

```
<video controls width="640">
    <source src="video.mp4" type="video/mp4">
    <source src="video.webm" type="video/webm">
    Your browser does not support video.
</video>
```

Output (In Browser)



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Hands-On

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Hands-On: Add Audio, Video & Comments

1 Add Audio

Add an audio player using the **<audio>** element and provide the appropriate source.

```
<audio controls>
    <source src="audio/sample.mp3" type="audio/mpeg">
    Your browser does not support the audio element.
</audio>
```

2 Add Video

Add a video player using the **<video>** element and provide the appropriate source.

```
<video controls width="400">
    <source src="video/sample.mp4" type="video/mp4">
    Your browser does not support the video element.
</video>
```

3 Add Comments

Add HTML comments explaining at least three sections of your code.

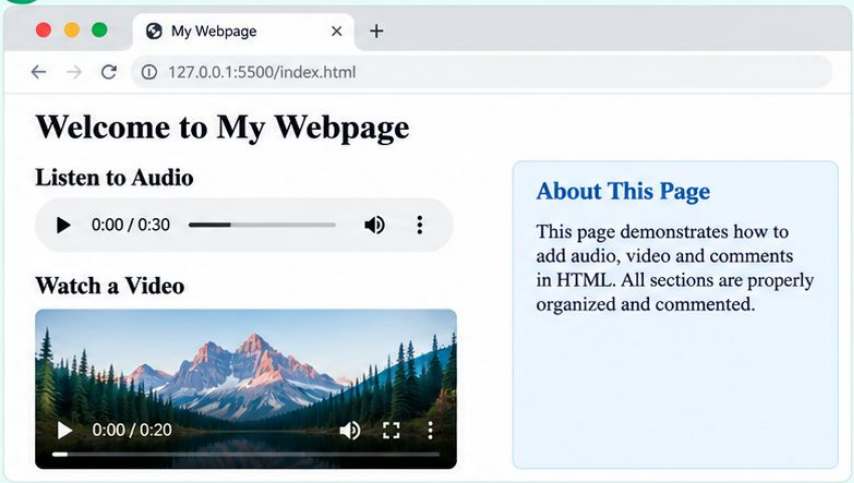
```
<!-- This is the header section -->
```

4 Code Organization

Ensure your code follows best practices:

- Proper indentation
- Proper nesting
- Correct closing tags
- Lowercase HTML tags
- Meaningful attribute names
- Useful comments
- Related content is logically organized

Sample Output (What your webpage should look like)



Code Example with Comments

```
<!-- Header section -->
<h1>Welcome to My Webpage</h1>

<!-- Audio section -->
<h2>Listen to Audio</h2>
<audio controls>
    <source src="audio/sample.mp3" type="audio/mpeg">
    Your browser does not support the audio element.
</audio>
```

```
<!-- Video section -->
<h2>Watch a Video</h2>
<video controls width="400">
    <source src="video/sample.mp4"
      type="video/mp4">
    Your browser does not support
    the video element.
</video>
```

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HTML Comments and Code Organization

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Why Use Comments?

HTML comments are notes written inside HTML code that are **not displayed on the webpage.**

Comment Syntax

```
<!-- This is an HTML comment -->
```

Example

```
<h1>My Website</h1>
<!-- This paragraph introduces
     the website -->
<p>Welcome to my website.</p>
```

The browser ignores the comment when displaying the page.

Output (In Browser)



Pro Tip: Use comments to explain your code, leave notes for yourself or others, or temporarily disable code without deleting it.

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HTML Code Organization and Best Practices

Best Practices

- **Use proper indentation** to show the structure of elements.
- **Use meaningful names** for attributes such as `id` and `class`.
- **Keep elements properly nested** and close elements correctly.
- **Use lowercase HTML tags** for consistent code style.
- **Use comments** when they help explain complex sections.
- **Keep related content together** and organize the document logically.
- **Use semantic elements** where appropriate instead of unnecessary generic elements.

Good Code Structure Example

```
<!DOCTYPE html>
<html lang="en">
  <head>
      <meta charset="UTF-8">
      <meta name="viewport" content="width=device-width, initial-scale=1.0">
      <title>My Website</title>
  </head>
  <body>
      <!-- Header Section -->
      <header id="main-header">
         <h1>Welcome to My Website</h1>
      </header>
      <!-- Navigation Section -->
      <nav class="navbar">
         <ul>
            <li><a href="#home">Home</a></li>
            <li><a href="#about">About</a></li>
            <li><a href="#contact">Contact</a></li>
         </ul>
      </nav>
      <!-- Main Content -->
      <main class="content">
         <section id="about">
            <h2>About Us</h2>
            <p>This is the about section of our website.</p>
         </section>
      </main>
      <!-- Footer Section -->
      <footer id="footer">
         <p>&copy; 2024 My Website. All rights reserved.</p>
      </footer>
  </body>
</html>
```

Pro Tip: Well-organized HTML is easier to read, maintain, and scale for larger projects.

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Example

Poorly Organized

```
<html><body><h1>My Website</h1><p>Welcome</p><p>Learn HTML</p></body></html>
```

Well Organized

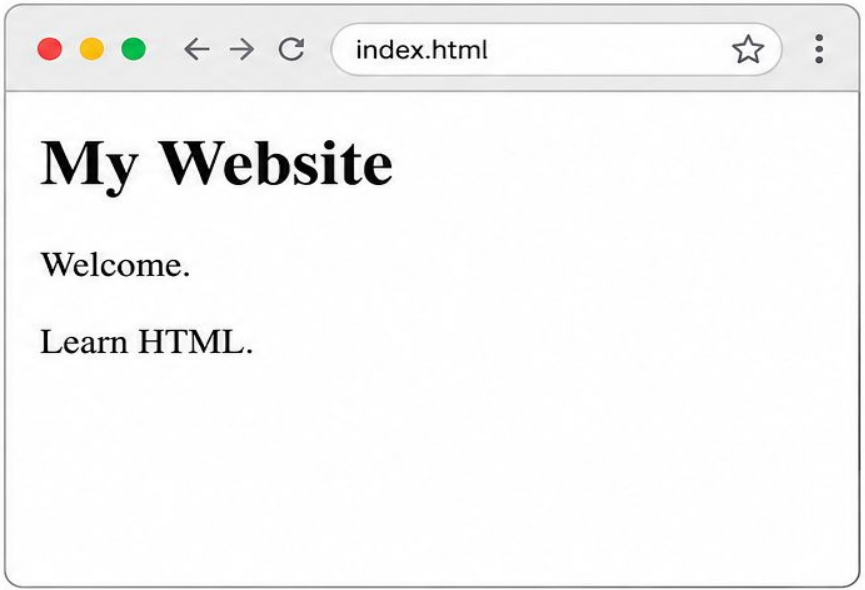
```
<html>
  <head>
    <title>My Website</title>
  </head>

  <body>
    <h1>My Website</h1>

    <p>Welcome.</p>
    <p>Learn HTML.</p>
  </body>

</html>
```

Output (In Browser)



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Thanks!


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